

Toluwani Samuel Aremu

AI Safety | Trustworthy AI | Responsible AI
[Certifications](#) • [Email](#) • [GitHub](#) • [Google Scholar](#) • [LinkedIn](#) • [Website](#)

RESEARCH INTERESTS

- Safe and trustworthy generative AI via technical control, alignment, and evaluation.
- Model provenance and attribution via robust watermarking techniques.
- Model safety and reliability through robust monitoring primitives and verification mechanisms.
- Threat modeling and analysis of advanced AI risks, including existential risk scenarios.
- Societal impacts of AI development and deployment.
- Alignment strategies grounded in philosophical and sociological theories of human behavior.
- AI Governance and Human-AI collaboration (e.g., AI debate, self-alignment, cooperative AI).

EDUCATION

- Machine Learning (PhD) | **Mohamed Bin Zayed University of Artificial Intelligence (MBZUAI), UAE** | (AUGUST 2023 – PRESENT)
 - **Advisor:** [Nils Lukas](#).
 - **Thesis:** [Towards Safe & Trustworthy Generative AI Systems](#).
 - **Expected Completion Date:** [May 2027](#).
- Machine Learning (MSc) | **Mohamed Bin Zayed University of Artificial Intelligence (MBZUAI), UAE** | (JANUARY 2021 – DECEMBER 2022)
- Computer Science (MSc) | **University of Ibadan (UI), Nigeria** | (MAY 2018 – MAY 2020)
- Computer Science (BSc) | **Adeleke University, Nigeria** | (OCTOBER 2012 – JULY 2016)
- Theory and Practice of Music (Diploma) | **Associated Board of the Royal Schools of Music (ABRSM), UK** | (AUGUST 2012 – AUGUST 2015)

RESEARCH EXPERIENCE

- Visiting Researcher | **Center for Frontier AI Research (CFAR), A*STAR** | (APRIL 2026 – PRESENT)
 - **Research Areas:** Watermarking for Generative AI.
 - **Host:** [Jie Zhang](#).
- Doctoral Researcher | **Spot-AI Lab, MBZUAI** | (AUGUST 2023 – PRESENT)
 - **Research Areas:** Watermarking for Generative AI, Alignment, AI Governance.
- Research Assistant | **MCR Lab, MBZUAI** | (FEBRUARY 2023 – AUGUST 2023)
 - **Research Areas:** Generative AI Applications.
- Graduate Research Assistant | **SPRINT-AI Lab, MBZUAI** | (JANUARY 2021 – JANUARY 2023)
 - **Research Areas:** Homomorphic Encryption, Secure Multi-Party Computation, Private Inference, Privacy-Preserving ML.

WORK EXPERIENCE

- Technical Co-Founder/Lead Quant | **Stealth (Incubated)** | (MAY 2025 – PRESENT)
 - Serving as the founding technical lead at an AI-powered trading startup incubated under [Kora](#).
- AI Engineer (Contract) | **Vadis Media, Switzerland** | (MAY 2025 – JULY 2025)
 - Led the core development (80%) of an AI-driven film production platform to connect producers to financiers and brands.
- Applied Scientist (Intern) | **M42, UAE** | (FEBRUARY 2023 – APRIL 2023)
- Data Analyst | **State Government of Osun, Nigeria** | (JANUARY 2018 – DECEMBER 2018)
- IT Support Engineer | **Government of Ekiti State, Nigeria** | (NOVEMBER 2016 – NOVEMBER 2017)
- Hardware Engineer (Intern) | **Lautech Teaching Hospital, Nigeria** | (JUNE 2015 – NOVEMBER 2015)
- Network Engineer (Intern) | **OAU-INTECU, Nigeria** | (JUNE 2014 – NOVEMBER 2014)

PUBLICATIONS (* denotes equal contribution, ** denotes working paper)

- **[Preprint, **]** Toyin, Hawau Olamide, Mutiah Apampa, **Toluwani Aremu**, Humaid Alblooshi, Ana Rita Valente, Gonçalo Leal, Zhengjun Yue, Zeerak Talat, and Hanan Aldarmaki. "Aligning Stuttered-Speech Research with End-User Needs: Scoping Review, Survey, and Guidelines." arXiv preprint arXiv:2604.20535 (2026).
- **[Preprint, **]** Aremu, Toluwani, Daniil Ognev, Samuele Poppi, and Nils Lukas. "Robust Safety Monitoring of Language Models via Activation Watermarking." arXiv preprint arXiv:2603.23171 (2026).
- **[CVPR-AdvML, *]** Fares, Samar, Klea Ziu, **Toluwani Aremu**, Nikita Durasov, Martin Takáč, Pascal Fua, Karthik Nandakumar, and Ivan Laptev. "Mirrorcheck: Efficient adversarial defense for vision-language models." The 6th Workshop of Adversarial Machine Learning on Computer Vision: Safety of Vision-Language Agents. 2026.
- **[Preprint, **]** Aremu, Toluwani, Noor Hussein, Munachiso Nwadike, Samuele Poppi, Jie Zhang, Karthik Nandakumar, Neil Gong, and Nils Lukas. "Mitigating Watermark Forgery in Generative Models via Randomized Key Selection." arXiv preprint arXiv:2507.07871 (2025).
- **[ICML, Spotlight, AR: 26.9%]** Diaa, Abdulrahman, **Toluwani Aremu**, and Nils Lukas. "Optimizing Adaptive Attacks against Watermarks for Language Models." In International Conference on Machine Learning, pp. 13546-13569. PMLR, 2025.
- **[Patent]** KANG, Wong Yu, **Toluwani AREMU**, Younes BALAH, Abdulmotaleb El Saddik, Maryam NADEEM, and Ivo Gollini NAVARRETE. "System and method for scanning faces and discovering scholarly information." U.S. Patent 12,423,369, issued September 23, 2025.
- **[AI Magazine (AAAI), AR: 24%]** Aremu, Toluwani, Oluwakemi Akinwehinmi, Chukwuemeka Nwagu, Syed Ishtiaque Ahmed, Rita Orji, Pedro Arnau Del Amo, and Abdulmotaleb El Saddik. "On the reliability of Large Language Models to misinformed and demographically informed prompts." Association for the Advancement of Artificial Intelligence (AAAI) AI Magazine 46, no. 1 (2025): e12208.
- **[ACL, Main, AR: 20.3%]** Nwadike, Munachiso S., Zangir Iklassov, **Toluwani Aremu**, Tatsuya Hiraoka, Benjamin Heinzerling, Velibor Bojkovic, Hilal AlQuabeh, Martin Takáč, and Kentaro Inui. "Library-Like Behavior In Language Models is Enhanced by Self-Referencing Causal Cycles." In Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers), pp. 25365-25377. 2025.

- [\[CVPR, Spotlight, AR: 22.1%\]](#) Vayani, Ashmal, Dinura Dissanayake, Hasindri Watawana, Noor Ahsan, Nevasini Sasikumar,..., **Toluwani Aremu** et al. "All languages matter: Evaluating Imms on culturally diverse 100 languages." In Proceedings of the Computer Vision and Pattern Recognition Conference, pp. 19565-19575. 2025.
- [\[IJCAI, AR: 14%\]](#) Tastan, Nurbek, Samar Fares, **Toluwani Aremu**, Samuel Horvath, and Karthik Nandakumar. "Redefining contributions: shapley-driven federated learning." In Proceedings of the Thirty-Third International Joint Conference on Artificial Intelligence, pp. 5009-5017. 2024.
- [\[SAI, Oral, AR: 37%\]](#) **Aremu, Toluwani**, Li Zhiyuan, Reem Alameeri, Mustaqeem Khan, and Abdulmotaleb El Saddik. "SSIVD-net: a novel salient super image classification and detection technique for weaponized violence." In Science and Information Conference, pp. 16-35. Cham: Springer Nature Switzerland, 2024.
- [\[Preprint\]](#) **Aremu, Toluwani**. "Unlocking Pandora's Box: Unveiling the Elusive Realm of AI Text Detection." Available at SSRN 4470719 (2023).
- [\[SaTML, Oral, AR: 26.3%\]](#) **Aremu, Toluwani**, and Karthik Nandakumar. "Polykervnets: Activation-free neural networks for efficient private inference." In 2023 IEEE Conference on Secure and Trustworthy Machine Learning (SaTML), pp. 593-604. IEEE, 2023.

OTHER EXPERIENCE

- **Projects**
 - Accented Speech Recognition | **Course Project (MBZUAI)** | (AUGUST 2021 – DECEMBER 2021)
 - Racial Bias Mitigation in Self-Supervised Face Recognition Architecture | **Course Project (MBZUAI)** | (JANUARY 2021 – MAY 2021)
 - Gender Bias Mitigation in Word Embeddings | **Course Project (MBZUAI)** | (JANUARY 2021 – MAY 2021)
- **Teaching**
 - Teaching Assistant, Introduction to AI (AI1010) | **MBZUAI** | (AUGUST 2025 – DECEMBER 2025)
 - Teaching Assistant, Mathematical Foundations of AI (MTH701) | **MBZUAI** | (AUGUST 2024 – DECEMBER 2024)
 - Teaching Assistant, Object Oriented Programming (OOP) | **University of Ibadan** | (JAN 2019 – MAY 2019)
- **Talks/Presentations**
 - Mitigating Watermark Forgery Attacks in Generative Models via Multi-Key Watermarking | **MBZUAI Research Showcase, UAE** | (OCTOBER 2025)
 - Literature Review and Research Methodologies | **LyngualLabs, Nigeria** | (MAY 2025)
 - Optimizing Adaptive Attacks against Content Watermarks for Language Models | **ICLR-WMARK Workshop, Singapore** | (APRIL 2025)
 - "Ethical Perspectives of AI", presented at the **Department of Material Sciences, University of Denver, USA**. (JUL 2022)
- **Service (Reviewer/Program Committee/Area Chair/Program Chair)**
 - **Conferences:** [CVPR](#) (2026) | [NeurIPS](#) (2025, 2026) | [ICLR](#) (2025, 2026) | [ICML](#) (2025, 2026) | [AISTATS](#) (2025) | [AAAI](#) (2025, 2026) | [DLJ](#) (2025)
 - **Workshops:** [HRAIM@NeurIPS](#) (2025) | [SafeGenAI@NeurIPS](#) (2025) | [WMark@ICLR](#) (2025)
 - **Journals:** [IEEE Access](#) (2025) | [OSJ](#) (2025) | [CHBAH](#) (2025) | [AI Magazine](#) (2025) | [IEEE TPAMI](#) (2025)
- **Leadership/Volunteering**
 - Resident Life Assistant | **MBZUAI Undergraduate Student Life** | (AUGUST 2025 – MAY 2026)
 - Volunteering Member | **EVOLVE-MBZUAI** | (AUGUST 2024 – DECEMBER 2025)
 - Volunteering Member | **Spruce Up Your Space** | (AUGUST 2024 – FEBRUARY 2026)
 - Associate Editor | **ML Department Research Blog** | (JANUARY 2024 – PRESENT)
 - Student Editor | **MBZUAI Research Blog** | (JANUARY 2024 – PRESENT)
 - Mentor | **Graduate School Research Mentorship for Africans** | (JANUARY 2021 – PRESENT)

HONORS & AWARDS

- MBZUAI Conference Travel Scholarship. (APR 2025, JUL 2025)
- MBZUAI PhD Fully Funded Fellowship. (AUG 2023)
- MBZUAI MSc Fully Funded Fellowship. (JAN 2021)
- UAE Golden Visa for Talented Persons/Specialists in Science. (OCT 2022)
- MBZUAI Award of Appreciation for Iconic Representation and Student Hospitality. (AUG 2022)
- ProjectSet Innovation Challenge for Entrepreneurship (ICE-22). (MAY 2022)
- Top 100, DeepLearning.AI Data-Centric AI Competition. (AUG 2021)
- NYSC-FRSC Award for the Most Creative Corp Member. (OCT 2017)
- 2015 AUE-NACOSS Award for the Best Programmer. (JUL 2015)

REFERENCES

- Available on request